

DOCUMENT 02

Research findings and code audit

What the references mean and what the current code actually supports

1. Reading the user's references



Desired: compact text-shaped footprint



Reference: rectangular overreach

User-provided conceptual references. The colored overlays are illustrative, not segmentation ground truth.

The desired reference uses a compact highlight around the upper-right sound effect and around the lettering in the lower balloon. The rectangular reference extends across surrounding art and the balloon boundary. The design goal is a text-shaped footprint with padding, not a slightly smaller rectangle.

The samples contain Korean lettering. They should be used as visual acceptance references, not fed into a benchmark as if the colored overlays were original text pixels. Obtain an unmarked source page and independently annotate removal masks for a meaningful model test.

Interpretation: disconnected components are allowed. The lower text can have a compact grouped footprint where a small margin connects nearby letters. No requirement has been established to erase all white space inside a word or paragraph envelope.

Important limit: these supplied images are illustrative. This report does not claim they show the actual masks emitted by the installed app, nor that either image was produced by a tested model.

2. Verified current architecture

The following findings describe publicly inspected source, not a reproduced desktop run.

Area	Source finding	Consequence for the new option
CTD adapter	Uses the bounding-box head and one-channel segmentation output; the two-channel line map is not used. Its box classes are weak language labels. [S02]	Add line-map recovery as a separately measured experiment. Never require a language class for the no-OCR policy.
Ogkalu support	Existing balloon helpers adopt uncovered text and seed masks, including an Otsu-style fallback. [S04]	Extend that candidate mechanism; do not reinstall an already-present capability. Audit fallback mask safety.
Mask representation	The Rust Mask stores binary membership inside rectangular storage bounds. [S07]	Arbitrary shapes are already representable. A rectangle can remain the storage extent without becoming the edit area.
Fitting	Fitted carries mask, ink and paper masks. Automatic growth and manual-authoritative paths differ. [S06]	Add an explicit geometry policy; do not overload a “manual” flag for automatic text-shaped regions.
Engine geometry	Model application grows the ink mask with an isolation margin; LaMa uses the narrower ink path. [S08, S09]	In the new mode, account for every effective margin before preview and commit.
Fill	Fill works with the fitted region and sampled paper. [S10]	Separate sampling extent from write support so fill cannot widen the approved footprint.
Compositor	Writes only through a patch's applied mask. [S11]	Reuse this invariant and ensure the applied mask is exactly the previewed support.

Policy findings

The documented automatic workflow gates bubble text to Japanese and holds outside-bubble text for review by default. The outside-bubble “Clean anyway” setting is separate from engine selection. [S05, S19]

The run's model readiness logic currently includes gate files. A real no-recognition policy therefore needs mode-aware capability checks and model initialization, not only an OCR checkbox. [S15, S18]

Do not conflate: not detected; detected but held; rejected by language policy; accepted but mask incomplete; and cleaned but inpainting poor. Each needs its own diagnostic outcome.

3. Why the overlay and the edit can disagree

Frontend finding. RegionLayer.svelte positions region buttons using bbox percentages and draws rectangular outlines. That is a geometric display of region bounds, not a rendering of the pixel mask. [S16]

API finding. ApiRegion includes bbox and mask metadata, while the examined ApiMask representation describes a layer/provenance record rather than transmitting the authoritative pixel footprint to this overlay. Add a versioned preview-mask reference or equivalent route. [S14]

Existing protection. The compositor already enforces changed pixels as a subset of applied patch masks. The new feature should strengthen the UI/backend agreement around that existing protection, not bypass the compositor. [S11]

Specific places to inspect while implementing

Verified path	Proposed responsibility
crates/cleaner-core/src/detect/mod.rs	Candidate and segmentation output contracts.
crates/cleaner-core/src/detect/boxes.rs	Keep bbox geometry separate from edit membership.
crates/cleaner-core/src/fit/mod.rs	Explicit legacy versus text-shaped geometry policy.
crates/cleaner-core/src/mask.rs	Deterministic round dilation and bounded mask operations.
crates/cleaner-core/src/engines/model.rs	New-mode write support, alpha and removal of hidden post-preview growth.
crates/cleaner-core/src/engines/lama.rs	Consume the prepared mask plan, not infer a new one.
crates/cleaner-core/src/engines/fill.rs	Read sampling context; write only through approved support.
crates/cleaner-core/src/composite.rs	Final enforcement and outside-mask regression tests.
src-tauri/src/run.rs	Policy-aware readiness, discovery and preview lifecycle.
src-tauri/src/library.rs	Versioned geometry metadata and mask preview transport.
src/lib/editor/RegionLayer.svelte	Draw actual text-shaped support and retain accessible selection.

These are change targets, not a completed patch. Additional denoise, manual-tool, export and state call sites must be traced in the implementation checkout.

4. Research conclusion: segmentation is the missing link

Baseline: use the CTD mask more effectively

Upstream CTD processes its line map, refines predicted masks against source-image evidence, and has an optional path for segmentation outside known blocks. The refinement combines candidate color/threshold components guided by the prediction. [S20, S21]

Recommendation: port or adapt the concept into a separately testable Rust refinement stage. Do not blindly copy all hole filling, dilation or thresholds: the new policy must control those operations and avoid double padding. Whole-component acceptance can leak into artwork when lettering touches an illustration.

Additional segmentation candidate: SAM-TS / Hi-SAM

This project explicitly supports pixel-level text-stroke segmentation as well as broader word, line and paragraph masks. The stroke output is the relevant candidate; a paragraph mask is not a safe substitute. Its reported evaluations are on scene-text datasets, not this Korean comic sample. [S27, S28]

Recommendation: compare a base-sized stroke checkpoint on difficult local crops. Its checkpoints require the corresponding SAM image encoder, and the documented workflow is a Python research stack. Do not assume the full model is a ready-made ONNX plugin. [S27]

Comic-specific alternative

The ogkalu YOLOv8m segmenter is described as trained on about 3,000 comic/webtoon images. The short model card does not establish cleanup-mask granularity or superiority to CTD. Inspect exported masks before choosing it. [S26]

What the research supports

Build the new mask option independently of a detector upgrade. Refined CTD is the first baseline; a specialized segmenter is a measurable optional enhancement. CTD + COO can address discovery, while segmentation and controlled compositing address rectangular overreach.

Research limitations: main/master may change; no immutable app commit or installed-build identity was verified. No models were executed, no synthetic score is presented, and no app source was modified. Source identifiers resolve in Document 08 and the per-document source links.

Source IDs link directly to primary references. Full evidence ledger: Document 08.